The AI Earth Observation and Data Playbook

How AI turns photons into products people bet money, crops and rescues on - without ever mistaking a detected pixel for a verified fact.



Gates EO0-EO5 | Eight templates | 20 golden rules

A detected pixel is not a verified fact. Earth observation AI is licensed per product, per sensor, per region and per use - never in general, and never on a demo scene.

The planet, photographed daily; the truth, still earned scene by scene

Volumes 1 to 3 licensed the mission, the design and the fleet. This volume licenses what the fleet is for: the data. Earth observation is where space AI meets its largest market and its largest temptation, because a satellite image looks like evidence. It has the authority of a photograph and the provenance of a model output, and the gap between those two things is where this volume lives.

AI is genuinely strong here. It can search a petabyte archive in seconds, detect change across a continent overnight, and turn raw downlinks into maps before the next pass. It is also genuinely dangerous here, for a structural reason: its errors are geographic. A misclassified field, a missed flood, a phantom vessel - each lands on a real place, owned by real people, who never consented to be a training example.

The claim ladder

- **Rung 1 - Captured.** The photons arrived, were downlinked and archived with provenance intact.
- **Rung 2 - Calibrated.** Radiometry and geometry are corrected against known references, to a stated tolerance.
- **Rung 3 - Analysed.** A model has produced a detection, classification or measurement, with uncertainty attached.
- **Rung 4 - Verified.** The product has been checked against independent ground truth, per region and season, and passed.
- **Rung 5 - Transferable.** The verified performance holds on a new sensor, region or season without re-earning it - claimed rarely, granted almost never.

The Earth observation rule: **every product is sold, shared or acted on at the rung it has actually earned - per product, per sensor, per region and per use.**

EXECUTIVE LENS

Fund verification, not visualisation.

MISSION LENS

A beautiful map of a wrong answer is the most expensive artefact in the building.

OPERATIONS LENS

If ground truth is not in the budget, the product is not in the catalogue.

Ten stages from tasking to decision, graded stage by stage

The chain below grades each stage for AI licence at EO4, the highest working level short of scale. The pattern matches Volume 1: AI is strongest where checking is cheap - search, triage, change flagging - and weakest where the output becomes a claim about the world that someone will act on.

Stage	What AI does	Licence
1. Tasking	Rank collection requests, resolve conflicts	EO4 - inside signed fences
2. Ingest and archive	Catalogue, quality-flag, deduplicate	EO4 - full
3. Calibration	Correct radiometry and geometry	Deterministic-led, AI-assisted
4. Screening	Cloud, haze and artefact masking	EO4 - full, masks preserved
5. Detection	Find objects and change	EO4 - with uncertainty
6. Classification	Label land, water, crop, structure	EO4 - per region and season
7. Measurement	Areas, counts, heights, indices	EO4 - verified products only
8. Fusion	Combine sensors and external data	EO4 - provenance per layer
9. Alerting	Notify customers of events	EO4 - thresholds human-set
10. Decision triggers	Act automatically on a product	Refused at EO4 - human-signed

Stage 10 is the line. An alert invites a human decision; a trigger replaces one. Payouts, evacuations, interdictions and trades that fire directly from a model output are not licensed in this volume at any accuracy, because accuracy is a property of a test set and a trigger fires in the world.

EXECUTIVE LENS

The map is the portfolio: fund the weak stages, not just the demonstrable ones.

MISSION LENS

Stage 10 is where liability concentrates; keep it human.

OPERATIONS LENS

A stage graded low is a boundary, not a backlog item.

The scarcest pixel is the one you pointed at the wrong place

Every image begins as an argument about the future: this place, at this time, will matter. Tasking AI ranks thousands of collection requests against capacity, weather probability, customer priority and licence value - and it is excellent at it, provided the fences are signed by humans first.

- **Priority is a policy, not a prediction.** Disaster tasking outranks commercial tasking because a human policy says so, never because a model inferred urgency from a customer's tone.
- **Conflicts are surfaced with prices.** When two customers want the same pass, both see the trade and its cost; the system never silently drops the smaller account.
- **Speculative capture is capped.** Imaging on the chance of a sale consumes power, downlink and archive forever; the cap is a budget line, reviewed monthly.
- **Standing refusals.** No tasking that breaches an imaging restriction, a licence condition or a privacy exclusion zone (Chapter 14), however it is phrased in the request.

Tasking decisions are logged with their reasons, because the most common EO dispute is not about a bad image - it is about the image that was never taken, on the day everything happened.

EXECUTIVE LENS

Tasking is portfolio allocation with a shutter.

MISSION LENS

Pre-sign the disaster priority; do not negotiate it during the flood.

OPERATIONS LENS

Log the collections you declined; they are tomorrow's hardest questions.

Optical flatters, radar persists, and every sensor lies in its own dialect

A model trained on one modality knows one way of seeing. Licensing is per sensor because the failure modes are per sensor, and a product that silently mixes modalities inherits the worst failure of each.

Modality	Sees	Characteristic AI failure
Optical (visible / multispectral)	Colour, texture, vegetation indices	Cloud shadow read as water; seasonal colour read as change
Synthetic aperture radar	Structure and moisture, day / night, through cloud	Speckle read as texture; layover in terrain read as objects
Hyperspectral	Material signatures, chemistry	Signature drift with atmosphere; library mismatch sold as detection
Thermal infrared	Heat, fire, industrial activity	Sun-heated surfaces read as activity; emissivity ignored
Lidar / altimetry	Height, canopy, elevation	Sparse sampling interpolated into false precision
RF / signals-derived	Emitters, vessel beacons	Absence of signal sold as absence of activity

Fusion (Chapter 9) can cover a modality's blind spot, but only when each layer keeps its own provenance and uncertainty. A fused product that cannot say which sensor contributed which claim cannot be verified, and what cannot be verified stays at rung 3.

EXECUTIVE LENS

Buy sensors for the failure you cannot tolerate, not the brochure image.

MISSION LENS

Per-sensor licensing feels slow until the first cross-sensor incident.

OPERATIONS LENS

Absence of signal is a finding about the sensor, not the world.

Rung 2 is earned against the physical world, not against last month's output

Calibration is the least glamorous stage and the one every later claim stands on. Radiometric calibration ties pixel values to physical radiance using onboard references, lunar and desert calibration sites, and cross-checks against peer satellites. Geometric calibration ties pixels to places. AI assists both - flagging drift, predicting degradation - but the references themselves are physical, maintained, and never generated.

Ground truth is infrastructure

- **Reference sites are assets.** Instrumented fields, buoys, gauges and survey plots are funded like ground stations, because without them every accuracy figure is a rumour.
- **Truth expires.** A validation site describes the season and year it was measured in; land changes, and stale truth verifies nothing.
- **Independence is the point.** Ground truth collected by the team being graded, using the product being graded, is not ground truth - it is a mirror.
- **Hold some truth back.** A sealed validation set, never used in training or tuning, is the only honest final exam a product ever sits.

Calibration rule: **a product's accuracy claim may never exceed the quality, currency and independence of the ground truth beneath it.**

EXECUTIVE LENS

Ground truth is the cheapest insurance in the entire chain.

MISSION LENS

When calibration slips, every downstream product slips silently with it.

OPERATIONS LENS

Protect the sealed validation set like a password.

From photons to products, every level keeps its receipts



Figure 1 - One scene, five levels: raw capture unrolls into calibration, correction, classification and a decision product, each layer keeping the receipts of the one below.

The standard ladder of processing levels is also a ladder of claims, and each level must remain reproducible from the one below. The moment a pipeline overwrites its inputs, every product it ever made becomes unverifiable at a stroke.

Level	Content	Rule
L0	Raw downlink, as received	Immutable, archived, never edited
L1	Calibrated, geolocated imagery	Deterministic processing; AI flags, never alters
L2	Surface products: reflectance, temperature, masks	Algorithm and version recorded per scene
L3	Composites, mosaics, time series	Every contributing scene listed; gaps shown as gaps
L4	Analysis products: detections, maps, measurements	Model, training data and uncertainty recorded

Cloud and quality masks are data, not cosmetics. A composite that fills a cloudy week with a model's plausible guess has manufactured evidence; gap-filling is permitted only when the filled pixels are labelled as filled, in the product and in every export of it.

EXECUTIVE LENS

Reproducibility is the product; the map is a by-product.

MISSION LENS

Never let a pipeline eat its own inputs.

OPERATIONS LENS

A labelled gap is honest data; a silent fill is a lie with good manners.

Change is a claim about two dates, and both of them can be wrong

Change detection is EO's flagship AI product and its most quietly fragile. A change map compares a before and an after, and every difference between those scenes that is not the change of interest - season, sun angle, sensor drift, a different satellite - arrives in the output disguised as change.

- **Pair discipline.** Change is computed only between scenes matched for sensor, season window and geometry, or the mismatch is modelled and its uncertainty carried into the product.
- **Report the base rate.** In a stable landscape, most flagged change is false. Precision at the landscape's real base rate - not at the balanced test set's - is the only number a customer may be quoted.
- **Negative change matters.** What disappeared - forest, water, structures - is usually the decision-relevant half, and the half demo scenes never show.
- **Persistence filters are policy.** Requiring change to persist across three passes removes noise and delays truth; the delay is a product decision, signed, not a tuning detail.

Every change product ships with its confusion profile by land-cover class. A single headline accuracy for a change map is treated in this volume the way a single average for a fleet was treated in Volume 3: as a place where errors go to hide.

EXECUTIVE LENS

Sell change with its base rate attached.

MISSION LENS

The before image deserves the same audit as the model.

OPERATIONS LENS

Delay is a decision; sign it like one.

A label without a confidence is a rumour in a uniform

Classification turns pixels into categories - crop, water, building, burn scar - and categories into money: subsidies, premiums, permits, fines. That is why this volume's rule for classification is stricter than for detection. A detection says something is there; a classification says what it is, and downstream systems will treat the label as fact unless the product physically prevents them.

- **Uncertainty travels with the label.** Per-pixel or per-object confidence is part of the product format, not a sidecar file nobody opens. Products that strip uncertainty on export are recalled.
- **Abstention is a feature.** A model licensed to say unknown, and routed to human review when it does, outperforms a model forced to guess - on every metric that involves consequences.
- **Classes are contracts.** The definition of forest, damage or occupancy used by the model is written down, versioned and shown to the customer; silent redefinition between versions is a product recall event.
- **Error is priced by class.** Confusing wheat with barley costs money; confusing water with land during a flood costs lives. Acceptance thresholds are set per class, by consequence.

EXECUTIVE LENS

Price errors by consequence, never by frequency alone.

MISSION LENS

An abstaining model is a safer product and a better business.

OPERATIONS LENS

If the class definition is not written, the product is not finished.

Analysis-ready means the arguments have already happened, honestly

Analysis-ready data (ARD) promises that imagery from different satellites and sensors arrives co-registered, calibrated and comparable, so customers can analyse instead of cleaning. AI does much of the alignment and harmonisation - and each harmonisation choice is a small editorial decision about the world that must stay visible.

- **Harmonise the measurements, never the conclusions.** Aligning grids and units is processing; smoothing away a disagreement between two sensors is censorship of evidence.
- **Disagreement is a layer.** Where optical and radar products differ, the ARD stack carries both and a disagreement flag. The flag is often the most valuable layer in the stack - it marks exactly where a human should look.
- **Lineage per pixel.** Every ARD cell can answer: which scenes, which models, which versions, which masks. If lineage stops at the product level, verification stops there too.
- **External data is a guest.** Weather, cadastral and crowdsourced layers enter with their own provenance and error models, and never silently upgrade an EO claim's rung.

Fusion rule: **combine sensors to cover blind spots; never average away a disagreement that was trying to tell you something.**

EXECUTIVE LENS

ARD sells trust; lineage is what makes it true.

MISSION LENS

The disagreement layer is a feature customers will pay for.

OPERATIONS LENS

A guest dataset cannot raise the host product's rung.

The flood does not wait for the validation report, so the licence is written before the rain



Figure 2 - The analyst and the responder: inside the amber ring, flood extent verified against ground reports; outside it, model shading stays grey and unacted-on.

Disaster response is EO's finest hour and its hardest licensing problem: products must move in hours, ground truth is underwater, and the cost of a wrong map is measured in misdirected rescues. The answer is not to abandon verification - it is to pre-verify. Disaster products are licensed in peacetime, per hazard and per region type, with degraded-mode rules signed in advance.

- **Two-speed products.** A fast product, labelled provisional, issued inside hours for situational awareness; a verified product follows, and every provisional map carries its expiry and its known failure modes on its face.
- **Pre-signed thresholds.** Evacuation-support and resource-dispatch thresholds are agreed with response agencies before the event, so no one negotiates statistics mid-crisis.
- **Absence is reported as absence.** No coverage, cloud-blocked and not-yet-processed are three different statements, and conflating them with no damage is the classic disaster-data failure.
- **After-action truth.** Every disaster product is graded afterwards against surveyed damage, and the grade feeds the licence - including when it lowers it.

EXECUTIVE LENS

Provisional is a licence class, not an apology.

MISSION LENS

Write the degraded-mode rules on a quiet day.

OPERATIONS LENS

Grade every disaster product afterwards, especially the heroic ones.

Slow truths, long baselines, and the discipline of not improving the past

Climate products invert the disaster problem: nothing is urgent and everything is permanent. A thirty-year record of ice, forest or sea-surface temperature is a scientific instrument, and AI reprocessing of the archive is where its integrity is most at risk.

- **The record is reprocessed whole, or not at all.** Applying a better model to recent years only manufactures a trend at the seam. Every algorithm change triggers a full-archive reprocessing and a published seam analysis.
- **Version the planet.** Each release of a climate data record is immutable and citable; corrections create a new version, never a quiet edit to the old one.
- **Uncertainty grows honestly with age.** Older sensors were worse; the record says so, in numbers, rather than letting a model polish early decades to modern smoothness.
- **Claims discipline.** Attribution - this change was caused by that activity - is a higher rung than detection of change, and is licensed separately, with the causal evidence named.

Environmental enforcement products - illegal clearing, discharge, encroachment - carry an additional duty: they will be contested, in court and in public, by parties with resources. Chain of custody and reproducibility (Chapter 6) are what let a true product survive a hostile audit.

EXECUTIVE LENS

A manufactured trend is worse than a missing decade.

MISSION LENS

Reprocess whole or publish the seam; there is no third option.

OPERATIONS LENS

Build enforcement products to survive the audit of an opponent.

A field is someone's livelihood rendered in pixels

Agricultural EO - yield estimation, crop classification, irrigation and stress mapping - is the largest commercial EO market and the clearest case for per-region, per-season licensing. A model that reads Iowa maize superbly can fail silently on smallholder plots an order of magnitude smaller, in a different intercropping pattern, under a different sky.

- **Plot size is a licence boundary.** Accuracy is certified at a stated minimum mapping unit; below it, the product abstains rather than averaging a village into a field.
- **Season is a licence boundary.** Performance is certified per phenological stage; a model verified at canopy closure is unverified at emergence, whatever the annual average says.
- **Aggregation hides harm.** Regional yield can be right while every individual farm estimate is wrong - and it is the individual estimate that sets the loan, the premium and the payout.
- **Ground campaigns are shared infrastructure.** Crop-cut surveys and farm visits, run with local partners and paid for, are the ground truth this market stands on (Chapter 5).

Agriculture rule: **if a product touches an individual's livelihood, it is verified at the individual's scale - or it does not touch them.**

EXECUTIVE LENS

Regional accuracy can coexist with individual injustice; audit both.

MISSION LENS

Smallholder contexts are a different product, not a harder demo.

OPERATIONS LENS

Pay for ground campaigns; they are the market's foundations.

Watching the world includes being watched back

Vessel detection, port activity, infrastructure monitoring and construction verification are commercially powerful and dual-use by nature: the same product that catches illegal fishing can track a navy's logistics, and the same model that counts trucks for a retailer can count them for an adversary.

- **Dark targets are a claim, not a colour.** A vessel not broadcasting, found by radar or optical detection, is reported with detection confidence and the known false-alarm profile - harbours generate phantom ships in radar the way deserts generate phantom lakes in optical.
- **Customer screening is part of the product.** Who may buy which resolution, latency and area coverage is a licensing decision reviewed like export control, because functionally it is one.
- **Latency is a capability.** A monthly mosaic and a fifteen-minute alert are different weapons and different products, licensed separately even when the pixels are identical.
- **Reciprocity.** Competitors and states operate equivalent systems over your own facilities; security planning (Volume 8) assumes your sites are in someone else's change-detection queue.

Standing refusal: no product whose purpose is identifying or tracking named private individuals, and no sale into a use the seller would not defend in public, on the record, with the product sheet attached.

EXECUTIVE LENS

Dual-use is not an edge case; it is the business model, governed openly.

MISSION LENS

Latency and resolution are capabilities - license them as such.

OPERATIONS LENS

The public-defence test settles most hard sales calls in one meeting.

Resolution is a policy decision that physics keeps renegotiating

Modern EO does not resolve faces or number plates - and that sentence has a shelf life. Resolution, revisit and AI enhancement improve together, and super-resolution models can manufacture detail that was never captured, which is worse than capturing it: a hallucinated feature in an enhanced image is evidence invented by software.

- **Enhancement is labelled, always.** Any AI-enhanced or super-resolved product states what was captured and what was inferred, at the product level and in the metadata. Selling inferred detail as captured imagery is fraud, and treated as such.
- **People-adjacent uses get human review.** Products over homes, schools, places of worship and protests - encroachment, activity, gatherings - are reviewed per use, however coarse the pixels.
- **Imaging restrictions are honoured in the tasking system** (Chapter 3), not in a policy PDF: shutter-control rules, national restrictions and customer-specific exclusions are enforced by the software that schedules the satellite.
- **Retention is a decision.** Archives of high-cadence imagery over populated areas are a surveillance asset in waiting; retention periods are set, published and audited.

EXECUTIVE LENS

Publish your retention and enhancement rules before a journalist asks.

MISSION LENS

Enforce restrictions in the scheduler, not in a memo.

OPERATIONS LENS

Inferred detail sold as captured fact ends companies.

Who owns a picture of everyone's planet

An EO product can involve the satellite operator, the tasking customer, the analytics provider, the landowner imaged, and a public archive - and each may hold a different right in the same pixel. Licensing chaos downstream is the direct result of rights vagueness upstream, so this volume treats the licence as part of the product specification.

- **Derived-product rights are written before training.** Whether a customer's tasked imagery may train models that serve their competitors is answered in the contract, in plain words, before the first scene is collected - not discovered in a dispute.
- **Open data is a strategy, not a leak.** Public missions demonstrated that free medium-resolution data grows the entire market for paid high-resolution and analytics products. The open tier is chosen deliberately, with its licence (attribution, share-alike, no-derivatives) stated per product.
- **Government tasking carries public-interest terms.** Disaster imagery bought with public money carries pre-agreed release rights for responders and the public, negotiated in peacetime (Chapter 10).
- **Attribution survives export.** Licence and provenance metadata are embedded so that a product separated from its portal still states its source, rung and permitted uses.

EXECUTIVE LENS

Rights vagueness upstream becomes litigation downstream.

MISSION LENS

Decide the training question in the contract, while everyone is still friendly.

OPERATIONS LENS

Open tiers grow markets; unplanned openness just loses them.

Models learn the places you showed them, and fail in the places you did not

EO training data has a geography, and it is not the Earth's. Labelled examples cluster where it is cheap to label: wealthy, cloud-free, well-mapped regions, photographed in their dry seasons. Models inherit that map, and their global accuracy claims quietly describe the labelled world, not the world.

- **Publish the training map.** Every product sheet shows where its training and validation examples come from - regions, seasons, sensors - next to the accuracy claim they support.
- **Label quality is measured, not assumed.** Inter-annotator agreement is audited per label campaign; labels drawn on imagery of a different season than the scene they describe are a known, priced error source.
- **Local knowledge is expertise.** Annotators who know the landscape - its crops, boats, roofs and seasons - are specialists, recruited and paid as such; generic labelling of unfamiliar terrain manufactures confident nonsense.
- **Synthetic training data is fenced.** Simulated imagery may augment rare classes, but a product validated only against synthetic scenes has validated against its own assumptions (Volume 2's twin discipline, applied to data).

Training data rule: **an accuracy claim extends exactly as far as the geography of the evidence beneath it - and not one field further.**

EXECUTIVE LENS

The training map belongs on the product sheet, beside the price.

MISSION LENS

Pay local annotators as the specialists they are.

OPERATIONS LENS

Synthetic validation is a mirror, not a window.

Rung 4 is a per-region, per-season, per-class verdict - or it is marketing

Validation is where EO products earn rung 4, and where most quietly fail to. A single global accuracy figure, measured once, on a convenience sample, in one season, is not validation; it is advertising with decimals. This chapter sets the standard the gates enforce.

Dimension	Validated separately	Why
Region	Each biome / market region in the licence	Landscapes differ more than models do
Season	Wet and dry, growing and dormant	The same field is four different classification problems
Class	Every decision-bearing class	Averages bury the class that matters
Sensor state	Per satellite and after any anomaly	Calibration drift is a silent model change
Time	Re-validated on a rolling cadence	The world drifts; last year's rung 4 expires

Validation uses the sealed ground-truth set (Chapter 5), sampled independently of the product team, and results are published internally in full - including the regions that failed, which are then removed from the licence rather than footnoted. A licence that shrinks on evidence is the system working; a licence that never shrinks is a system that has stopped measuring.

EXECUTIVE LENS

Publish failures internally in full, or they will be published externally in part.

MISSION LENS

Shrinking licences are evidence of life.

OPERATIONS LENS

Expiry dates on accuracy claims, like on food.

A perfect product delivered late is a historical document

EO value decays at different rates: a flood map in hours, a crop map in weeks, a climate record in years - but incorrectly delivered value can be worse than late value. Delivery is part of the product's licence, because format and channel decide whether uncertainty, provenance and expiry survive to the decision.

- **APIs carry the whole product.** Uncertainty, masks, lineage, rung and licence terms are fields in the API response, not a PDF beside it. A customer integration that cannot ingest the uncertainty field is flagged at onboarding, not discovered at the incident review.
- **Latency tiers are licensed tiers.** Provisional-fast and verified-slow are separate products with separate prices and labels (Chapter 10), never the same feed at two speeds.
- **Offline and low-bandwidth delivery are designed,** because disaster zones and rural customers - often the highest-value users - have the worst connectivity. Tiled, resumable, small-first delivery is an engineering requirement, not a courtesy.
- **Corrections propagate.** When a product is corrected or recalled, every delivered copy is reachable: version notices go to every API key and download record, automatically.

EXECUTIVE LENS

If the API drops the uncertainty, the product is not delivered - a rumour is.

MISSION LENS

Corrections must travel as fast as the original error did.

OPERATIONS LENS

Design delivery for the bandwidth of the flood zone, not the office.

Deciding in orbit saves downlink and spends reversibility

Processing on the spacecraft - screening clouds before downlink, detecting events onboard, prioritising the downlink queue - moves AI upstream of the last reversible moment. On the ground, a bad product can be withdrawn; in orbit, a scene discarded by an onboard model is gone at the speed of the pass.

- **Onboard models advise the queue; they rarely delete.** Licensed pattern: onboard AI ranks and compresses, with a signed sampling rule guaranteeing a fraction of rejected scenes still downlink for audit. Deletion policies require EO4 evidence per scene type.
- **The audit sample is sacred.** The randomly downlinked rejected scenes are how anyone knows what the onboard model is throwing away. When the sample shows drift, the onboard licence demotes automatically (Volume 1's demotion rule).
- **Uplinked models are flight software.** An onboard model update follows Volume 3's canary, soak and rollback discipline, per spacecraft - a fleet-wide silent model push is refused for the same reason a fleet-wide command push is.
- **Power and thermal are in the accuracy budget.** Onboard inference competes with the payload for watts; products state the operating envelope in which their onboard performance was verified.

EXECUTIVE LENS

Never let an unaudited model decide what the archive never sees.

MISSION LENS

The rejected-scene sample is small, cheap and priceless.

OPERATIONS LENS

An onboard model is flight software; treat its updates that way.

A geospatial foundation model is a supplier, with a supply chain

Large pre-trained geospatial models now sit beneath many EO products, and they import the licensing problem at scale: trained on whose imagery, labelled by whom, valid where, and changed when? The procurement questions are Volume 1's vendor discipline, sharpened for data products.

- **Training provenance is a contract item.** Sources, licences and the training map (Chapter 16) are disclosed for any model inside a licensed product; undisclosed provenance caps the product at rung 3.
- **Silent updates are configuration changes.** A vendor model version change re-opens validation for every dependent product (Chapter 17), at the vendor's cost of proof, not the customer's.
- **Benchmarks are scenes the vendor chose.** Evaluation happens on the buyer's sealed scenes, in the buyer's regions and seasons, before signature - the EO form of Volume 2's failure-seeking campaign.
- **Exit is designed.** Products must be re-hostable: features, embeddings and fine-tunes exportable, so a vendor exit degrades service rather than deleting the catalogue.

Vendor rule: **no provenance, no rung 4. A model that cannot say what it learned from cannot say where it works.**

EXECUTIVE LENS

Buy evaluation rights on your own scenes, not benchmark tours.

MISSION LENS

Price the exit before you price the licence.

OPERATIONS LENS

Undisclosed provenance is a cap, not a quibble.

Pixels are nearly free; trustworthy products are not

EO economics are seduced by a falling number - cost per image - that measures the wrong thing. Images are inputs. The unit a customer can use is a verified data product: a map, measurement or alert that earned rung 4 for its stated region, season and use. This volume's unit cost is therefore cost per verified data product, fully loaded.

Cost line	Counted
Tasking, downlink and archive	Per scene, including speculative capture
Processing and compute	Full chain, including reprocessing
Ground truth and validation	Campaigns, sealed sets, re-validation cadence
Human review	Abstention handling, disaster grading, audits
Corrections and recalls	Propagation, customer remediation, credits
Licensing and compliance	Rights, privacy review, export screening

Counted honestly, AI usually still wins - on triage, search, screening and change flagging, the savings are real. It wins by less than the pitch, and the honest number is the one that survives a customer's audit, a regulator's question and a bad season, all three of which are scheduled events in this industry.

EXECUTIVE LENS

Cost per image is a vanity metric; cost per verified product is a business.

MISSION LENS

Verification is a cost line to defend, not overhead to cut.

OPERATIONS LENS

The honest number survives three audits; the pitched number survives one meeting.

Anyone can pull a product; restarting it takes evidence and two signatures

The operating model mirrors the rest of the series, adapted to products rather than spacecraft. The unit of control is the product licence: product, sensor, region, season and use, recorded in one register. An unregistered product is not sold, shared or fed to another system - it does not exist, however good its demo was.

- **Halt authority is universal.** Any analyst, operator or customer-facing engineer can pull a product from delivery on a stated doubt. The pull is instant and blameless; the review that follows is neither.
- **Restart takes two signatures,** one from the product owner and one from validation, with the re-earned rung stated. Products restart at the rung the evidence supports, not the rung they held.
- **Demotion is automatic.** Calibration drift, a failed re-validation, a vendor model change or a disaster after-action grade below licence demotes the product without a meeting. Promotion always takes a meeting; demotion never waits for one.
- **The register is the catalogue.** Sales sell from the register, APIs serve from the register, and the register's rung is what appears on every product sheet.

Operating rule: **the register is the truth, the product sheet is a copy of it, and any gap between the two is an incident.**

EXECUTIVE LENS

Blameless pulls, rigorous restarts.

MISSION LENS

Automate demotion; deliberate promotion.

OPERATIONS LENS

If sales and the register disagree, stop the sale, not the register.

WORKED EXAMPLE - AURORA EARTH, PART 1

The audit: a global analytics claim meets nine regions and two seasons

Fictional worked example. All organisations, figures and results are illustrative, built to show the method, not to report real performance.

Aurora Orbital's Earth-observation arm sells tasking, imagery and analytics from the fleet licensed in Volume 3. An analytics vendor claims its platform cuts processing and analyst cost by 48 percent, releases 61,000 hours a year, and delivers decision-ready global products - flood extent, crop classification and vessel detection - at verified quality, worldwide, from day one.

What the audit found

- **Savings.** 48 percent claimed verified at 19 percent. 26,400 hours genuinely released, 16,800 banked; the rest was review work the pitch had classified as eliminated and the products still required.
- **Flood extent.** Claimed rung 4, worldwide. Verified at rung 4 in three of nine licensed regions - and at rung 2, calibrated only, in wet-season monsoon terrain, where radar layover and inundated vegetation defeated the model the demo had never met.
- **Crop classification.** Verified at rung 4 for large-plot temperate regions in-season; smallholder regions dropped to rung 3, analysed-not-verified, until local ground campaigns (Chapter 12) could be run.
- **Vessel detection.** Strong in open ocean; harbour false-alarm rate five times the quoted figure, exactly as Chapter 13 predicts.

Nothing in the audit suggested a bad vendor. It suggested a global claim wrapped around regional truths - the standard shape of EO overclaim, and the reason licensing is written per region in the first place.

EXECUTIVE LENS

Global is a claim, not a region.

MISSION LENS

The audit found a good product wearing an oversized licence.

OPERATIONS LENS

Wet season is a different planet; price it that way.

WORKED EXAMPLE - AURORA EARTH, PART 2

EO5 granted in part: the licence that survived was smaller and worth more



Figure 3 - The Aurora Earth operations wall: every product cell shows its rung, and the quarantined cell in red is a product pulled on a calibration doubt, pending re-validation.

The EO4 licence was written around the evidence: tasking, ingest, screening and change flagging licensed fleet-wide; flood extent licensed at rung 4 for three regions with two-speed provisional delivery (Chapter 10) elsewhere; crop classification licensed per region and season window; vessel detection licensed for open ocean with harbour zones routed to human review. Cost per verified data product settled at 0.84 against a 0.98 pre-AI baseline - a real, bankable improvement, roughly half the pitch.

Refused

- **Automated insurance payout triggers** from crop and flood products - stage 10 of the map, refused at any accuracy.
- **Worldwide flood marketing** until wet-season validation is earned region by region.
- **AI super-resolution enhancement** sold as captured detail (Chapter 14).

EO5 scale was granted in part: the licensed cells scale without new evidence; expansion into new regions, seasons or uses returns to EO1. Twelve months on, the register shows four licence expansions earned, two refused, and one product demoted after a calibration drift - each recorded as the system working, because a register that only grows is a register nobody is checking.

EXECUTIVE LENS

A smaller licence that holds beats a global one that folds.

MISSION LENS

Refusals are part of the product; customers trusted the register more, not less.

OPERATIONS LENS

Expansions earned: four. Refused: two. Demoted: one. That is what health looks like.

One page, to be read before any product ships

1. Every product carries a rung - captured, calibrated, analysed, verified, transferable - earned per product, per sensor, per region, per season and per use.

2. No product is sold, shared or acted on above its rung. Marketing language is audited against the register.

3. Processing is reproducible from immutable raw data; every level keeps algorithm, model and version receipts.

4. Uncertainty, masks and lineage travel with the product through every format, API and export.

5. Ground truth is independent, current and partly sealed; accuracy claims never exceed the truth beneath them.

6. Validation is per region, per season and per class, on a rolling cadence, with failures published internally in full.

7. Decision triggers are human-signed. Models alert; people decide; automation does not fire consequences.

8. Enhancement and gap-filling are labelled as inference, in the product and its metadata, without exception.

9. Privacy, imaging restrictions and customer screening are enforced in the tasking and delivery systems, not in memos.

10. Anyone can pull a product. Restart takes two signatures and re-earns its rung. Demotion is automatic on contrary evidence.

If a product cannot state its rung, its region, its season and its evidence in one sentence, **it is not ready to leave the building.**

EXECUTIVE LENS

Print it. The standard only works where people can see it.

MISSION LENS

Ten clauses, each one bought by somebody's incident.

OPERATIONS LENS

The one-sentence test closes most arguments.

SIX FAILURE PATTERNS

How Earth observation programmes actually go wrong

1. The demo scene

A product validated on the scenes that flatter it - clear skies, home regions, the good season - then sold on the average. Antidote: sealed validation on the customer's scenes (Chapters 5, 20).

2. The polished past

A new model reprocesses recent years only, manufacturing a trend at the seam, which is then reported as discovery. Antidote: whole-archive reprocessing and published seam analysis (Chapter 11).

3. The silent fill

Cloud gaps and missing passes are inpainted by a model and delivered as observation; decisions are made on weather that was invented. Antidote: gaps shown as gaps, fills labelled in every export (Chapter 6).

4. The average that drowned a region

Global accuracy holds while one region, season or class fails completely - usually the one the customer bought it for. Antidote: per-region, per-season, per-class licences (Chapter 17).

5. The trigger that fired itself

An alert quietly becomes an automated payout, dispatch or trade; the first public failure reveals nobody ever signed the decision. Antidote: stage 10 stays human-signed (Chapter 2).

6. The enhancement that invented a road

Super-resolution adds plausible detail; a planner, insurer or court treats invention as capture. Antidote: inference labelled as inference, always (Chapter 14).

EXECUTIVE LENS

Each pattern was a shortcut that looked like progress.

MISSION LENS

Patterns 3 and 6 are integrity failures, not accuracy failures.

OPERATIONS LENS

Teach the patterns; teams spot what they can name.

30 QUESTIONS

Asked at every gate, answered with evidence or not at all

- What rung has this product earned - per sensor, region, season and use?
- Where is the register entry, and does the product sheet match it?
- What ground truth sits beneath the accuracy claim, and when was it measured?
- Which validation set is sealed, and who holds it?
- What is the accuracy per class, at the real base rate?
- Which regions and seasons are excluded from the licence?
- How are cloud, gaps and fills labelled in the delivered product?
- Can every L4 product be reproduced from L0, with versions?
- Does uncertainty survive the API and every export format?
- When did calibration last drift, and which products demoted?
- What does the training map look like, beside the accuracy claim?
- Who labelled the training data, and what was their local knowledge?
- What fraction of onboard-rejected scenes is downlinked for audit?
- Which products are provisional, and who signed their thresholds?
- What changed in the vendor's model since validation?
- Can the catalogue be re-hosted if the vendor exits?
- Who may buy this resolution, latency and coverage - and who may not?
- What is the retention policy for imagery over populated areas?
- How are corrections propagated to every delivered copy?
- What is the cost per verified data product, fully loaded?
- Which savings are claimed, released and banked, separately?
- What is the abstention rate, and where do abstentions go?
- How are disaster products graded after the event?
- Which licence shrank on evidence this year - and if none, why not?
- Who can pull this product today, and have they ever?
- What would make this product demote automatically?
- Are enhancement and super-resolution labelled in the metadata?
- What rights does the customer have in derived and training uses?
- Which decision triggers exist, and whose signature is on each?
- If this product were wrong in public tomorrow, what evidence would we produce by noon?

A question answered with a dashboard is not answered; with the register and the sealed-set results, it is.

EXECUTIVE LENS

Question 30 is the one to ask in the room, out loud.

MISSION LENS

Unanswered questions are findings, not homework.

OPERATIONS LENS

Evidence or not at all - the gate rule for every volume in this series.

Six gates, each with the evidence that opens it

Gate	Decision	Evidence required
EO0 - Scope	Define products, rungs and exclusions	Product list with intended rung per region / use; standing refusals signed
EO1 - Prove	Calibration and validation stand up	Cal/val report; sealed-set results per region, season, class
EO2 - Pilot	Real customers, limited licence	Pilot register; abstention and correction logs; two-speed delivery working
EO3 - Integrate	Register drives catalogue and APIs	Uncertainty and lineage through every channel; correction propagation tested
EO4 - License	Operational sale at earned rungs	Per-product licences; screening and privacy enforcement in systems
EO5 - Scale	Expand cells, volumes and markets	Rolling re-validation funded; demotion automation live; unit cost verified

First 90 days

- **Days 1-30.** Build the register; assign every live product a rung or pull it; freeze global accuracy claims.
- **Days 31-60.** Seal a validation set; run per-region, per-season grading; label all fills and enhancements in delivery.
- **Days 61-90.** Move the register into the catalogue and APIs; test correction propagation; hold the first EO4 gate on evidence.

EXECUTIVE LENS

Gates open on evidence, not on enthusiasm.

MISSION LENS

The 90 days end with a smaller, truer catalogue.

OPERATIONS LENS

EO5 expands cells; it never widens claims by itself.

What an auditor, a customer or a court will eventually ask for

The evidence pack, per product

- Register entry: rung, sensor, region, season, use, expiry.
- Calibration history and drift log for every contributing satellite.
- Validation report: sealed-set results per region, season and class, with failures in full.
- Training map and label-quality audit; vendor model provenance and version log.
- Processing lineage: L0 to L4, algorithms, models, masks and fill labelling.
- Correction and recall log, with propagation receipts per delivered copy.
- Disaster after-action grades, where the product is hazard-related.

Sector annex - where the emphasis moves

Sector	Emphasis
Insurance and finance	Individual-scale verification; triggers stay human-signed; basis risk disclosed
Government and defence	Customer screening, latency licensing, chain of custody for contested products
Agriculture and food	Plot-size and season boundaries; local ground campaigns; smallholder abstention
Climate and environment	Whole-archive reprocessing; versioned records; enforcement-grade reproducibility
Energy and infrastructure	Change detection pair discipline; dark-target confidence; reciprocity planning
Media and public information	Enhancement labelling; provisional disaster labelling; no invented detail, ever

EXECUTIVE LENS

Assemble the pack before anyone asks; afterwards is too late and it shows.

MISSION LENS

The annex moves emphasis, never the standard.

OPERATIONS LENS

Basis risk disclosed is a product feature in insurance, not a weakness.

EIGHT FILL-IN TEMPLATES

The licence, made paperwork

Template 1 - Product Licence Card

Product: ___ | Sensor(s): ___ | Region / season: ___ | Use: ___ | Rung: ___ | Exclusions: ___ | Expiry: ___ | Signatures: ___

Template 2 - Validation Plan

Product: ___ | Sealed set holder: ___ | Regions / seasons / classes: ___ | Ground truth and date: ___ | Thresholds per class: ___ | Failure handling: ___

Template 3 - Tasking Fence

Priority policy: ___ | Disaster override: ___ | Speculative capture cap: ___ | Restriction zones enforced: ___ | Conflict pricing shown to: ___

Template 4 - Processing Lineage Record

Scene IDs (L0): ___ | Calibration version: ___ | Models (L2-L4): ___ | Masks: ___ | Filled pixels labelled: ___ | Reproduced by / date: ___

Template 5 - Disaster Product Release

Hazard / event: ___ | Class: provisional or verified | Expiry: ___ | Failure modes: ___ | Thresholds signed by: ___ | After-action grade due: ___

Template 6 - Product Pull and Restart

Product: ___ | Pulled by / reason: ___ | Customers notified: ___ | Restart evidence: ___ | Rung on restart: ___ | Two signatures: ___

Template 7 - Vendor Model Intake

Model / version: ___ | Provenance disclosed: ___ | Training map: ___ | Sealed-scene results: ___ | Update terms: ___ | Exit tested: ___

Template 8 - Verified Product Workbook

Period: ___ | Fully loaded chain cost: ___ | Verified products delivered: ___ | Cost per verified data product: ___ | Claimed / released / banked: ___

Scope, proof, tasking, lineage, release, recall, intake and cost - each short enough to complete during a flood, which is when several of them will be used. Paperwork from logs is evidence; from memory, fiction.

The volume in twenty lines, and the series in ten

- License per product, per sensor, per region, per season and per use.
- A detected pixel is not a verified fact.
- Accuracy never exceeds the ground truth beneath it.
- Validate per region, per season and per class - or not at all.
- Uncertainty, masks and lineage travel with the product.
- Gaps are shown as gaps; fills are labelled as inference.
- Enhancement is labelled, always; inferred detail is never sold as captured.
- Models alert; people decide; triggers stay human-signed.
- Reprocess the whole archive, or publish the seam.
- Publish the training map beside the accuracy claim.
- Keep a sealed validation set, and let licences shrink on evidence.
- Audit what onboard models throw away.
- Vendor model changes are configuration changes.
- Corrections travel as fast as the original error.
- Provisional disaster products carry expiry and failure modes on their face.
- Grade every disaster product after the event.
- Screen customers, resolution and latency like export controls.
- Count cost per verified data product, fully loaded.
- Anyone can pull a product; restart takes two signatures and re-earns its rung.
- A smaller true licence beats a global claim that folds.

AI in Space and Orbital Operations - the series

Vol	Title	Gates	Status
1	Space Strategy	SS0-SS5	Built
2	Mission Design and Autonomy	MD0-MD5	Built
3	Satellite and Constellation Operations	CO0-CO5	Built
4	Earth Observation and Data	EO0-EO5	Built - this volume
5	Conjunction, Debris and Orbital Safety	CD0-CD5	Next
6	Ground Systems and Mission Software	GS0-GS5	Planned
7	Lunar and Deep-Space Operations	LD0-LD5	Planned
8	Space Security and Resilience	SR0-SR5	Planned
9	Space Economics and Fleet Finance	EF0-EF5	Planned
10	Space Capstone and Assurance	SC0-SC5	Planned

Volume 5, Conjunction, Debris and Orbital Safety, is next. The series closes at ten with the Space Capstone and Assurance.